Spatial-Behavioral Analytics: Mapping Nutritional Vulnerability and Food Accessibility in Singapore's Aging Districts

ESTHER LEE YI SHAN, Nanyang Technological University, Singapore
YU ZHIYING ANGELA, Nanyang Technological University, Singapore
WOON YUN XUAN CLARA, Nanyang Technological University, Singapore
CHA JIA XUAN, Nanyang Technological University, Singapore

Singapore is ageing at exceptional velocity: by 2030 roughly one in four residents will be aged 65 or above, yet no published spatial index jointly captures the compounding effect of poor food access, weak public transit, and limited barrier-free infrastructure on elderly nutritional security. This paper presents a reproducible, end-to-end data mining pipeline that ingests eleven publicly available Singapore datasets at the Urban Redevelopment Authority (URA) subzone level, applies three corrected demographic and categorical filters including a composite wet-market type criterion (`TYPE IN ('MK','MHC')`), a minimum resident-population threshold of 10,000, and a dwelling-type socioeconomic proxy derived from Census 2020 HDB flat-type counts and executes three connected analytical phases across a validated set of residential subzones. Phase 1 runs Apriori association rule mining on a transparent thirteen-item binary matrix. Phase 2 compares K-Means, Agglomerative Hierarchical Clustering with Ward linkage, and DBSCAN using a mean rank protocol over silhouette, Davies-Bouldin, and Calinski-Harabasz metrics, and validates the chosen partition with a bootstrap Adjusted Rand Index stability analysis. Phase 3 trains a Random Forest against a documented proxy vulnerability target, benchmarks it against Logistic Regression, a depth-four Decision Tree, and a majority-class floor, and uses Gini feature importance to expose the dominant risk drivers. The corrected pipeline produces a ranked list of priority hotspot subzones concentrated in mature HDB precincts rather than low-density landed enclaves, with a three-layer evidence trail: co-occurrence rules, a stable neighbourhood typology, and a continuous vulnerability score. All outputs are regenerated from a single command in the public git repository.

CCS Concepts: • **Computing methodologies** → **Cluster analysis**; *Supervised learning by classification*; • **Human-centered computing** → Geographic visualization.

Additional Key Words and Phrases: food accessibility, elderly vulnerability, association rule mining, spatial clustering, random forest, Singapore, urban analytics

1 Introduction

1.1 Demographic and Urban Context

Singapore is at a critical demographic and urban inflection point. The number of citizens aged 65 and above is projected to grow from roughly 440,000 in 2015 to over 900,000 by 2030, meaning that one in four residents will soon be elderly [**?**]. The island's constrained land area of 715 km^2 and population density exceeding 7,797 persons per km^2 make this transition an acute spatial planning challenge. State frameworks the $3 billion Action Plan for Successful Ageing, the "City For All Ages" initiative, and the more recent "Age Well SG" programme [**?**] collectively champion the principle of ageing-in-place: transforming the built environment so that seniors can maintain daily independence within their own communities.

This aspiration is directly contingent on three spatial preconditions. First, affordable prepared food. Hawker centres, wet markets, and supermarkets are the primary food supply for older residents who cook less frequently and have lower average incomes than working-age cohorts; walking access to these venues is a daily necessity for residents who do not drive. Second, accessible public transit. The density of bus stops and MRT exits within a walkable radius determines whether residents with limited mobility can still reach food, healthcare, and social

Authors' Contact Information: Esther Lee Yi Shan, Nanyang Technological University, Singapore, Singapore, es0001an@e.ntu.edu.sg; Yu Zhiying Angela, Nanyang Technological University, Singapore, Singapore, yuzh0017@e.ntu.edu.sg; Woon Yun Xuan Clara, Nanyang Technological University, Singapore, Singapore; Cha Jia Xuan, Nanyang Technological University, Singapore, Singapore.

services. Third, barrier-free built infrastructure. Even where food venues and transit exist, steep approaches, absent lifts, and inaccessible building entrances can effectively exclude residents with limited mobility.

1.2 The Analytical Gap

Existing public indicators report each of these three dimensions in isolation. Traditional maps focus on one amenity at a time (e.g., only bus stops or only supermarkets). This approach hides the compounding effect of "amenity deserts," where multiple services are missing simultaneously. By ignoring how these deficits overlap, traditional mapping underestimates the true level of disadvantage in a neighbourhood [**?**]. In Singapore, current spatial indices fail to account for the compounding impact of multiple service deficits. When statistics are analysed in isolation, they mask the reality that a subzone lacking food access, transit, and barrier-free infrastructure simultaneously faces a much more severe disadvantage than one lacking only a single amenity. This lack of integrated mapping hides qualitatively different levels of hardship, preventing a true assessment of spatial inequality.

A second, subtler gap concerns the *population denominator*. Measuring elderly exposure by percentage share alone, without an absolute population floor, risks flagging ultra-low-density landed estates and parkland-fringe subzones as high-priority precisely because a handful of wealthy retirees inflates the percentage in a near-empty polygon. Likewise, measuring food proximity without distinguishing public-housing residents who are captive to pedestrian mobility from private landed-property residents, who rely on cars and domestic help, produces a spatially correct but sociologically inverted vulnerability map.

1.3 Research Questions

We address four concrete research questions.

RQ1. Which food, transit, and barrier-free indicators co-occur, or are jointly absent, across Singapore subzones at rates that exceed statistical independence?

RQ2. Can subzones be partitioned into distinct, stable neighbourhood types on the basis of elderly demographics, food access, and transit support?

RQ3. Which subzones are the highest priority for outreach under a transparent, documented proxy for nutritional vulnerability that controls for population size and socioeconomic dwelling type?

RQ4. Which features drive the vulnerability structure most strongly when it is recovered by a supervised classifier?

1.4 Contributions

Our main contributions are as follows.

- A fully reproducible pipeline that ingests eleven public datasets, applies three methodologically corrected filters composite wet-market typing, absolute population floor, and HDB dwelling-type proxy and produces all figures, tables, and model artefacts from a single command.
- A transparent thirteen-item binary matrix for Apriori mining in which every threshold is version-controlled and every item has a documented real-world meaning.
- A three-algorithm clustering comparison with a mean rank selection protocol and a bootstrap ARI stability analysis.
- A documented weighted z-score proxy vulnerability target whose formula, weights, and justifications are serialised alongside the numerical outputs, together with a four-model risk comparison exposing both the linear and rule-based structure of the target.
- A ranked hotspot list with a three-layer evidence trail that targets genuinely deprived public-housing precincts rather than low-density affluent enclaves.

2 Related Work

2.1 Urban Food Accessibility Measurement

The measurement of urban food accessibility has a long tradition in geography and public health. Classical buffer-based measures count food outlets within a fixed radius of each residential unit [?]. Gravity-style measures add a distance-decay function, weighting closer venues more heavily. More recent work moves to pedestrian network routing using OpenStreetMap data, which matters in dense Asian cities where expressways and drainage reserves create walking barriers invisible to straight-line calculations [?]. Our feature engineering uses centroid-to-venue Euclidean distances and one-kilometre buffer counts as proxies a necessary compromise given that network routing is computationally intensive at national scale but we acknowledge this as the primary engineering limitation and discuss its consequences in Section ??.

2.2 Association Rule Mining in Spatial Contexts

The Apriori algorithm [?] was originally designed for market-basket analysis. In spatial public health contexts it has been used to surface co-occurrence patterns among discrete indicators that pairwise correlation analysis cannot capture: a positive Pearson coefficient tells us two variables scale together, but it cannot quantify the rate at which all three of an elderly concentration, a food deficit, and a transit deficit appear in the same polygon simultaneously, nor can it express directional conditional rules. Our use case treats each subzone as a transaction represented by a binary indicator vector; the lift metric then quantifies excess co-occurrence relative to statistical independence.

2.3 Neighbourhood Typologies via Clustering

Clustering-based neighbourhood classification is a standard tool in geodemographics [?]. Internal validation metrics such as silhouette [?], Davies-Bouldin [?], and Calinski-Harabasz [?] are known to favour different configurations on the same data, so a robust selection protocol does not rely on any single metric. Density-based clustering via DBSCAN [?] adds qualitatively different geometric assumptions from the centroid-based K-Means and variance-based Ward algorithms. We include all three to avoid pre-committing to a particular cluster shape, and confirm the chosen partition with a bootstrap Adjusted Rand Index stability test [?].

2.4 Composite Vulnerability Indices

Composite vulnerability indices, including the CDC Social Vulnerability Index [?], typically take a weighted z-score form. The recurring criticism is that weights are opaque [?]. Our response is to serialise every weight, directional sign, and justification to a version-controlled configuration file and to write the resolved formula out as a plain-text artefact alongside the numerical scores. Critically, an earlier version of this pipeline also lacked an absolute population denominator, which caused affluent micro-population subzones to rank as highly vulnerable. We address this flaw explicitly in Section 3.3.

2.5 Random Forests and Interpretable Baselines

Random Forests [?] are ensembles of decision trees with bagging and random feature subsampling. Their robustness to feature scaling, tolerance of irrelevant features, and availability of Gini-based importance rankings make them attractive for tabular spatial data. A depth-constrained single decision tree is a natural interpretable counterpart, and Logistic Regression serves as a linear baseline to reveal how much of the proxy structure is linearly separable. We add a majority-class `DummyClassifier` as a sanity floor, following standard practice in applied machine learning [?].

Table 1. Eleven required datasets used in the pipeline. All files are loaded from `data/raw/`.

Dataset	Source	Role in the pipeline
Master Plan Subzone Boundary	URA / data.gov.sg	Spatial unit, 332 subzones
Census of Population 2020 (age)	SingStat	Elderly count and share
Census of Population 2020 (dwelling type)	SingStat	HDB flat-type counts for socioeconomic proxy
Hawker Centres (GeoJSON)	data.gov.sg	Affordable cooked food access
Supermarkets (GeoJSON)	data.gov.sg	Food retail access
NEA Market and Food Centre	NEA / data.gov.sg	Wet markets (`TYPE IN ('MK','MHC')`)
LTA Bus Stop	LTA / data.gov.sg	Bus transit access
LTA MRT Station Exit	LTA / data.gov.sg	Rail transit access
Friendly Buildings	BCA / data.gov.sg	Barrier-free infrastructure
Master Plan 2025 Land Use	URA / data.gov.sg	Residential land filter
Senior Activity and AAC Centres	data.gov.sg	Ageing support context

2.6 Socioeconomic Context in Spatial Vulnerability

A recurring finding in the food-desert literature is that spatial proximity to food venues is not sufficient to characterise vulnerability: financial affordability and transport mode choice mediate the relationship between distance and dietary outcomes [**?**]. In the Singapore context this is particularly acute because the built environment is sharply stratified by housing type. Residents of 1- and 2-room HDB rental flats are overwhelmingly reliant on pedestrian access and public transit and have household incomes well below the national median, whereas residents of private landed properties routinely use personal vehicles and domestic help for food logistics. A vulnerability model that treats all residents as equally dependent on nearby public amenities conflates two fundamentally different populations. We address this by incorporating a dwelling-type proxy from Census 2020 (Section 4.1.2).

3 Data

3.1 Datasets Used

The pipeline ingests eleven required datasets, all obtained from `data.gov.sg`, SingStat, or the LTA DataMall API. Table 1 summarises the full set. The eleventh dataset Census 2020 dwelling-type counts by subzone is new relative to the original pipeline and provides the HDB proxy described in Section 4.1.2.

Two optional origin-destination passenger volume files from the LTA DataMall API (`od_bus.csv` and `od_train.csv`) are supported but disabled by default. Constructing a defensible subzone-level mobility feature from these files requires decisions about stop-to-subzone geocoding, temporal aggregation, and node sharing that introduce researcher degrees of freedom beyond the scope of this study.

3.2 Data Cleaning and Filtering

3.2.1 Hawker Centres. The Hawker Centres GeoJSON contains a status field. We retain records whose status is one of `Existing`, `Existing (new)`, or `Existing (replacement)`, representing all centres currently in service.

3.2.2 Wet Markets: Corrected Type Filter. The NEA Market and Food Centre dataset classifies facilities using three distinct type codes: `HC` (pure Hawker Centre: cooked-food stalls only), `MK` (pure Market: fresh-produce stalls only), and `MHC` (combined Market and Hawker Centre: both fresh-produce and cooked-food stalls under one roof). Because the vast majority of neighbourhood wet markets in Singapore share a building with a hawker centre, they carry the `MHC` code rather than `MK`.

An earlier version of this pipeline applied the filter `TYPE == 'MK'`, which retained only three facilities island-wide. This was a critical schema misinterpretation: it silently discarded every co-located wet market in Singapore, artificially inflating the `nearest_market_km` feature for nearly all residential subzones and propagating a false scarcity of fresh-produce access through every downstream model. The corrected filter is `TYPE IN ('MK', 'MHC')`, which retains all facilities containing a fresh-produce market component regardless of whether a hawker centre is co-located.

3.2.3 Census Population. The Census 2020 age table is filtered to subzone-level rows. Elderly population is defined as residents aged 65 and above, summed across the six bands `Total_65_69` through `Total_90andOver`. Subzone names are normalised to title case with punctuation stripped before joining to the geometry.

3.2.4 Dwelling-Type Counts. The Census 2020 dwelling-type table provides counts of residents by flat type at subzone level. We extract the count of residents in 1-room and 2-room HDB rental flats (`HDB_1room` and `HDB_2room`) as well as the count of residents in private landed properties (`Landed_Private`). These counts are used to construct the HDB vulnerability proxy described in Section 4.1.2.

3.2.5 Land Use. The Master Plan 2025 Land Use layer is intersected with each subzone polygon. Four land-use type strings are recognised as residential: `RESIDENTIAL`, `RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY`, `COMMERCIAL & RESIDENTIAL`, and `RESIDENTIAL / INSTITUTION`.

3.3 Modelling Subset: Corrected Three-Stage Filter

Not all 332 subzones are realistic candidates for an elderly food vulnerability analysis. Industrial estates, military zones, water catchments, and the airport precinct have negligible resident populations and would distort both the clustering geometry and the score distribution. Beyond these morphological exclusions, two further demographic filters are required to prevent small-denominator fallacies from dominating the vulnerability ranking.

An earlier version of this pipeline retained a subzone if and only if its residential land share exceeded five per cent of its total area and its absolute residential area exceeded 5,000 m^2. While this successfully removed purely industrial subzones, it admitted micro-population enclaves landed estates, heritage bungalow zones, and parkland-fringe polygons with absolute resident counts below one thousand that are not representative of the public-housing elderly population for whom the pipeline is designed. Because the vulnerability proxy uses `elderly_pct` as a component, a handful of wealthy retirees in an otherwise empty polygon can generate a percentage score that dwarfs genuinely deprived precincts, producing a hotspot ranking that identifies affluent exclusivity rather than infrastructural deprivation.

The corrected modelling subset applies three conditions in sequence.

(1) **Morphological filter.** Residential land share $> 5\%$ of total subzone area and absolute residential area $> 5{,}000$ m^2. This removes purely industrial, military, and water-catchment subzones.
(2) **Demographic floor.** Absolute resident population $\geq 10{,}000$ persons (Census 2020). This removes micro-population enclaves such as Nee Soon (890 residents), Mount Pleasant (690 residents), Springleaf (4,230 residents), and Lakeside (Leisure) (1,070 residents), whose small denominators inflate `elderly_pct` to levels that override all other features in the vulnerability score.
(3) **Socioeconomic relevance filter.** Subzones where fewer than 20% of residents live in HDB flats of any type are flagged as private-landed dominant and are excluded from the priority hotspot ranking. Residents of private landed properties predominantly use personal vehicles and domestic help for food access; applying a pedestrian-proximity penalty to these subzones conflates infrastructure exclusivity with infrastructure deprivation.

Figure 1 shows the spatial result of the combined filters.

Fig. 1. Modelling subset after all three filters. Green subzones pass the morphological, demographic, and HDB-floor conditions. Grey subzones are excluded industrial or water zones. Navy-outlined polygons indicate micro-population enclaves removed by the demographic floor.

4 Methodology

The pipeline is organised into three analytical phases on top of a shared feature engineering layer. All spatial operations are performed in EPSG:3414 (SVY21), a projected coordinate reference system with units of metres. A global random seed (42) is set through a shared utility and passed explicitly to every scikit-learn estimator [**?**].

4.1 Feature Engineering

Feature construction is factored into six independent modules: `demographic`, `dwelling_type`, `food_access`, `transit`, `accessibility`, and `landuse`. A separate assembly module joins them on a normalised `subzone_key` into the canonical feature table.

4.1.1 Demographics. For each subzone i with resident population P_i, elderly count E_i, and area A_i in km^2:

$$\text{elderly_pct}_i = \frac{E_i}{P_i}, \quad \text{elderly_density}_i = \frac{E_i}{A_i}, \quad \text{elderly_density_log}_i = \log(1 + \text{elderly_density}_i). \tag{1}$$

The log transform is applied because elderly density is strongly right-skewed: a handful of high-density HDB precincts would otherwise dominate unscaled metrics.

4.1.2 HDB Dwelling-Type Socioeconomic Proxy. A key methodological addition in this paper is a socioeconomic weighting that prevents the vulnerability score from misidentifying private landed-estate subzones as deprived. Let R_i^1 and R_i^2 denote the counts of 1-room and 2-room HDB rental flat residents in subzone i, and let P_i be the total resident population. We define:

$$\text{hdb_low_pct}_i = \frac{R_i^1 + R_i^2}{P_i}. \tag{2}$$

This feature enters the vulnerability target formula with a positive sign, serving as a continuous proxy for the degree to which residents are dependent on pedestrian access to public food infrastructure. Subzones with near-zero `hdb_low_pct` but large Euclidean distances to hawker centres are correctly down-weighted in the vulnerability score.

4.1.3 Food Access. Let $\mathcal{H}$, $\mathcal{S}$, and $\mathcal{M}$ denote the sets of hawker centres, supermarkets, and wet markets under the corrected filter `TYPE IN ('MK', 'MHC')`. For each subzone centroid $\mathbf{c}_i$:

$$d_i^{(t)} = \min_{\mathbf{v} \in \mathcal{T}} \|\mathbf{c}_i - \mathbf{v}\|_2, \quad t \in \{H, S, M\}. \tag{3}$$

The count of each venue type within a 1 km buffer is $n_i^{(t)} = |\{\mathbf{v} \in \mathcal{T} : \|\mathbf{c}_i - \mathbf{v}\|_2 \leq 1000\,\text{m}\}|$. The composite food access score is the mean z-score of the three counts:

$$\text{food_access_score}_i = \frac{1}{3} \sum_{t \in \{H,S,M\}} z\big(n_i^{(t)}\big). \tag{4}$$

Food amenity diversity is the count of venue types with at least one venue in the buffer: $\text{food_amenity_diversity}_i = \sum_t \mathbf{1}[n_i^{(t)} \geq 1]$. The composite nearest-food distance is the minimum over the three types: $\text{nearest_food_km}_i = \min(d_i^{(H)}, d_i^{(S)}, d_i^{(M)})$.

Table 2. Engineered feature columns used in Phase 2 and Phase 3. `hdb_low_pct` is new relative to the original pipeline.

Feature	Dimension	High value means
`elderly_pct`	Demographics	Large elderly share
`elderly_density_log`	Demographics	Dense elderly per km^2
`hdb_low_pct`	Socioeconomic	High low-income HDB share
`nearest_hawker_km`	Food access	Far from nearest hawker
`nearest_supermarket_km`	Food access	Far from nearest supermarket
`nearest_market_km`	Food access	Far from nearest wet market
`food_amenity_count_log`	Food access	Many food venues nearby
`food_amenity_diversity`	Food access	Diverse venue types nearby
`transit_access_score`	Transit	Good bus and MRT access
`accessibility_support_score`	Accessibility	Many barrier-free buildings nearby

Fig. 2. Marginal distributions of the ten modelling features across the corrected residential subzone subset. Log-transformed features are approximately symmetric; nearest-distance features retain moderate right skew. `hdb_low_pct` shows a right-skewed distribution reflecting the concentration of low-income HDB stock in a subset of mature estates.

Fig. 3. Pearson correlation matrix of the ten modelling features. Transit access and food access are positively correlated, reflecting Singapore's transit-oriented development pattern. `hdb_low_pct` is positively correlated with elderly density and negatively correlated with nearest-food distances, consistent with the concentration of mature HDB estates near town centres.

All proximity features use Euclidean distance in EPSG:3414 rather than pedestrian network distance. In subzones bounded by expressways or drainage reserves, true walking distances exceed the Euclidean proxy; the impact of this approximation and its remediation paths are discussed in Section **??**.

4.1.4 Transit Access. Bus stops within 500 m and MRT exits within 800 m of the centroid are counted as b_i and m_i, log-transformed, and z-scored:

$$\text{transit_access_score}_i = \tfrac{1}{2}\big[z(\log(1+b_i)) + z(\log(1+m_i))\big]. \tag{5}$$

4.1.5 Barrier-Free Accessibility Support. The count of BCA-certified Friendly Buildings within 500 m, f_i, is log-transformed and z-scored:

$$\text{accessibility_support_score}_i = z(\log(1+f_i)). \tag{6}$$

This feature is a point-density proxy for barrier-free infrastructure. The underlying dataset provides building coordinates and addresses but no pedestrian network polylines, so a high score indicates accessible buildings are nearby but does not guarantee a continuous, barrier-free walking route between them. We flag this as an ecological fallacy in Section **??**.

4.1.6 Feature Summary. Table 2 lists the ten columns entering the clustering and risk model, including the new `hdb_low_pct` feature. Figure 2 shows marginal distributions and Figure 3 shows the pairwise Pearson correlation matrix.

Table 3. Binary items in the Apriori matrix. $Q_p(x)$ denotes the p-th percentile of feature x across the corrected modelling subzones.

Item	Condition	Meaning
`high_elderly_share`	`elderly_pct` $\geq Q_{0.66}$	Top third by elderly share
`high_elderly_density`	`elderly_density_log` $\geq Q_{0.66}$	Top third by density
`high_hdb_low_pct`	`hdb_low_pct` $\geq Q_{0.66}$	Top third by low-income HDB share
`poor_food_access`	`nearest_food_km` $\geq Q_{0.66}$	Far from any food venue
`rich_food_access`	`food_access_score` $\geq Q_{0.66}$	Many venues nearby
`poor_transit_access`	`transit_access_score` $\leq Q_{0.33}$	Weak transit
`rich_transit_access`	`transit_access_score` $\geq Q_{0.66}$	Strong transit
`poor_accessibility_support`	`accessibility_support_score` $\leq Q_{0.33}$	Few barrier-free buildings
`rich_accessibility_support`	`accessibility_support_score` $\geq Q_{0.66}$	Many barrier-free buildings
`hawker_present`	$n_i^{(H)} \geq 1$	At least one hawker within 1 km
`supermarket_present`	$n_i^{(S)} \geq 1$	At least one supermarket within 1 km
`market_present`	$n_i^{(M)} \geq 1$	At least one wet market within 1 km
`diverse_food_environment`	diversity ≥ 2	Two or more venue types nearby

4.2 Phase 1: Apriori Association Rule Mining

4.2.1 Binary Item Matrix. Apriori requires a Boolean transaction matrix. We construct an $N \times 13$ matrix by applying quantile-based thresholds to the continuous features, where N is the size of the corrected modelling subset. Every item is derived from one feature column and one threshold stored in `config/settings.yaml`. Table 3 defines the thirteen items. Item thresholds are computed after all three filtering stages to prevent distributional contamination from micro-population outliers. The item `high_hdb_low_pct` is new; it enables the detection of rules that link socioeconomic deprivation with food and transit deficits, a class of compound co-occurrence invisible to the original twelve-item matrix.

4.2.2 Algorithm and Parameters. For a binary transaction matrix $\mathbf{T} \in \{0,1\}^{N \times M}$ the Apriori algorithm derives rules $X \Rightarrow Y$ with $X \cap Y = \emptyset$ meeting minimum thresholds on:

$$\text{support}(X \cup Y) = \frac{|\{i : X \cup Y \subseteq t_i\}|}{N}, \tag{7}$$

$$\text{confidence}(X \Rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}, \tag{8}$$

$$\text{lift}(X \Rightarrow Y) = \frac{\text{confidence}(X \Rightarrow Y)}{\text{support}(Y)}. \tag{9}$$

Lift greater than 1 indicates co-occurrence beyond statistical independence. We use the `mlxtend` implementation [**?**] with min_support = 0.05, min_confidence = 0.50, min_lift = 1.10, and maximum itemset length four. We repeat the run at five support levels $\{0.03, 0.05, 0.08, 0.10, 0.15\}$ as a sensitivity analysis, and additionally conduct a discretisation robustness check by repeating the run at 50th, 66th, and 75th percentile bin cuts to confirm that the compounding-barrier rules do not depend on arbitrary threshold placement.

4.3 Phase 2: Unsupervised Clustering

4.3.1 Preprocessing and Algorithms. All ten feature columns are standardised with a `StandardScaler` (zero mean, unit variance) before fitting. Three algorithms are compared.

Table 4. Components of the operational vulnerability score V_i. All weights and signs live in `config/settings.yaml`. `hdb_low_pct` is new relative to the original pipeline.

Feature	Dir.	Wt.	Justification
`elderly_pct`	+	1.00	Higher elderly share concentrates vulnerability.
`elderly_density_log`	+	0.50	Density amplifies local demand for amenities.
`hdb_low_pct`	+	0.75	Low-income HDB residents are pedestrian-dependent; high share raises effective risk.
`nearest_food_km`	+	1.00	Greater walking distance to affordable food.
`food_access_score`	−	1.00	More nearby venues reduce effective risk.
`transit_access_score`	−	0.75	Transit enables reaching distant food.
`accessibility_support_score`	−	0.50	Barrier-free buildings reduce physical barriers.

K-Means minimises the within-cluster sum of squares:

$$\arg\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|_2^2, \tag{10}$$

with 25 random initialisations.

Agglomerative with Ward linkage merges the pair of clusters whose union minimises the increase in total within-cluster variance.

DBSCAN classifies a point as a core point if at least `min_samples` points lie within distance ε. We estimate ε from the k-distance elbow heuristic and sweep `min_samples` over $\{4, 6, 8\}$.

4.3.2 Metric Sweep and Selection Protocol. K-Means and Agglomerative are swept over $k \in \{2, \ldots, 8\}$. Each configuration is scored on silhouette, Davies-Bouldin, and Calinski-Harabasz. We rank every configuration by each metric separately and select the configuration with the lowest mean rank, avoiding over-reliance on any single criterion.

4.3.3 Bootstrap Stability. In each of 30 iterations a random 80% subsample is drawn without replacement and clustered independently with the chosen algorithm and k. The Adjusted Rand Index between successive pairs is recorded; the mean and standard deviation serve as a stability score [**?**].

4.4 Phase 3: Proxy Target and Risk Model

4.4.1 Operational Vulnerability Target. Because no validated external vulnerability label exists at the subzone level for Singapore, we construct a transparent operational proxy. The score is a weighted sum of standardised components:

$$V_i = \sum_{c \in C} \text{sign}(c) \cdot w_c \cdot z(x_{ic}), \tag{11}$$

where x_{ic} is the raw feature value, $z(\cdot)$ is the population z-score, $w_c \geq 0$ is the component weight, and $\text{sign}(c) \in \{+1, -1\}$ encodes whether a higher raw value increases or decreases vulnerability. The components are listed in Table 4. Relative to the original pipeline: `nearest_market_km` now reflects the corrected wet-market set, substantially reducing inflated distance values; and `hdb_low_pct` enters with positive sign and weight 0.75, ensuring that subzones with very few low-income HDB residents receive a lower effective score even if they are geographically distant from public food venues.

Subzones in the top 25th percentile of V_i receive the binary label $y_i = 1$. The resolved formula, weight configuration, and percentile cut-off are written to `outputs/tables/vulnerability_target_definition.txt`.

We are explicit that because y_i is a deterministic function of $\mathbf{x}_i$, Phase 3 is a structural recovery experiment rather than a traditional out-of-sample prediction task. The purpose is to verify that the Random Forest recovers the proxy structure, to compare its behaviour with simpler baselines, and to surface which features carry the most information through a Gini importance ranking.

4.4.2 Experimental Protocol. The modelling subzones are split into training (75%) and test (25%) sets using a stratified split that preserves the 3:1 class ratio. We report both held-out test-set metrics and stratified five-fold cross-validation metrics on the full dataset.

4.4.3 Models. **Random Forest (RF).** An ensemble of 400 decision trees, each fitted on a bootstrapped sample with a random feature subset of size $\sqrt{p}$ at each split. Balanced class weights, minimum leaf size of two, no maximum depth.

Logistic Regression (LR). L2 regularisation with balanced class weights; features standardised inside a `Pipeline` to prevent leakage; maximum 2,000 iterations.

Decision Tree (DT). A single tree constrained to maximum depth four with balanced class weights, producing at most 15 internal nodes that can be read directly.

Dummy (majority). `DummyClassifier` with `strategy = most_frequent`, providing a sanity floor.

4.4.4 Evaluation Metrics. We report five test-set metrics:

$$\text{Acc} = \frac{TP+TN}{N}, \quad \text{Prec} = \frac{TP}{TP+FP}, \quad \text{Rec} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R},$$

together with ROC-AUC. Cross-validation reports mean and standard deviation of accuracy, F_1, and ROC-AUC across five folds.

4.5 Reproducibility

Python 3.11 or 3.12 is required. Dependencies are pinned in `requirements.txt` and mirrored in `environment.yml` for Conda users. A `Dockerfile` and `docker-compose.yml` support full environment reproduction. All tunable parameters live in `config/settings.yaml`. Running `python run.py` regenerates every figure, table, joblib model, and pipeline log. A stage flag (`features`, `apriori`, `clustering`, `all`) allows partial runs. A unit test suite in `tests/` covers configuration loading, feature builders, Apriori rule generation, clustering sweep, and the risk model on a deterministic fixture.

5 Experiments and Results

5.1 Effect of Corrected Filters on the Modelling Subset